

AI-Based SEO Blog Post Creation Tool

Final Project Report

This project focuses on building an end-to-end AI-based automation system that generates SEO-optimized blog posts by analyzing trending products from e-commerce platforms. The goal is to reduce manual effort in content creation while improving organic traffic and scalability for businesses.

1. Objective

The objective of this project is to automate the complete SEO blog creation process including product research, keyword extraction, content generation, and publishing using Python and FastAPI.

2. Problem Statement

Traditional blog creation involves manual product research, keyword planning, content writing, and publishing, which is time-consuming and costly. Businesses struggle to scale SEO content efficiently.

3. Proposed Solution

The proposed system automatically scrapes trending products, extracts SEO keywords using NLP techniques, generates a 150–200 word SEO-friendly blog, and publishes it on CMS platforms.

4. System Architecture

The system follows a modular pipeline: Product Scraper → SEO Keyword Analyzer → AI Blog Generator → Publisher. Each module is independent and scalable.

5. Technologies Used

- 1 Python – Core programming language
- 2 FastAPI – Backend API framework
- 3 BeautifulSoup – Web scraping
- 4 TF-IDF – SEO keyword extraction
- 5 WordPress REST API – Blog publishing

6. Implementation Details

Trending products are scraped from an e-commerce demo website. SEO keywords are extracted using TF-IDF. The blog generator creates structured SEO content and saves or publishes it automatically.

7. Results & Impact

The system reduced blog creation time from hours to minutes. It ensures consistent SEO quality and allows businesses to scale content generation without additional manpower.

8. Use Cases

- 1 Digital marketing agencies
- 2 E-commerce companies
- 3 Startups and SaaS businesses
- 4 SEO freelancers

9. Conclusion

This project demonstrates how AI and automation can transform digital marketing workflows. The tool is scalable, cost-effective, and production-ready.